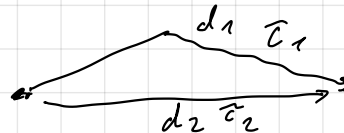


Wireless Receivers

Fading:



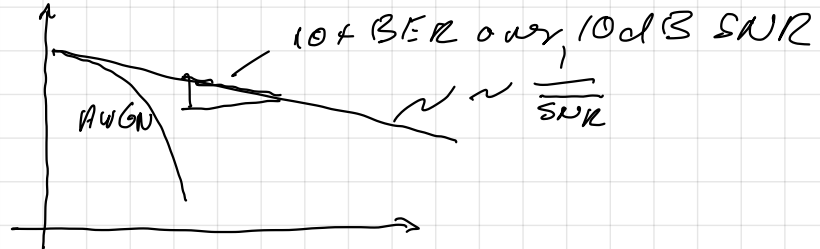
$$\frac{1}{N} \sum_{n=1}^N \frac{s(t - \tau_n)}{s(t)} e^{j\phi_n} = s(t) \underbrace{\sum_{n=1}^N e^{j\phi_n}}_{\underline{h}}$$

$$h: \mathcal{CN}(0, 1)$$

$$h_r: \mathcal{N}(0, 1/2)$$

$$h_i: \mathcal{N}(0, 1/2)$$

BER:



Optimum Receiver

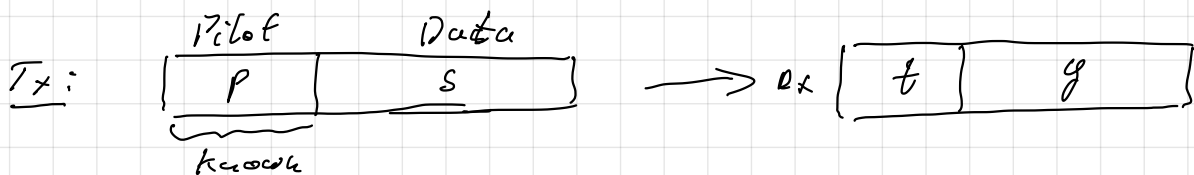
$$\underline{y} = \underline{h}s + \underline{w} \rightarrow \text{Receiver}$$

$\rightarrow$ Assume we know $\underline{h}$

$$\hat{s} = \arg \min_s \|\underline{y} - \underline{h}s\|^2$$

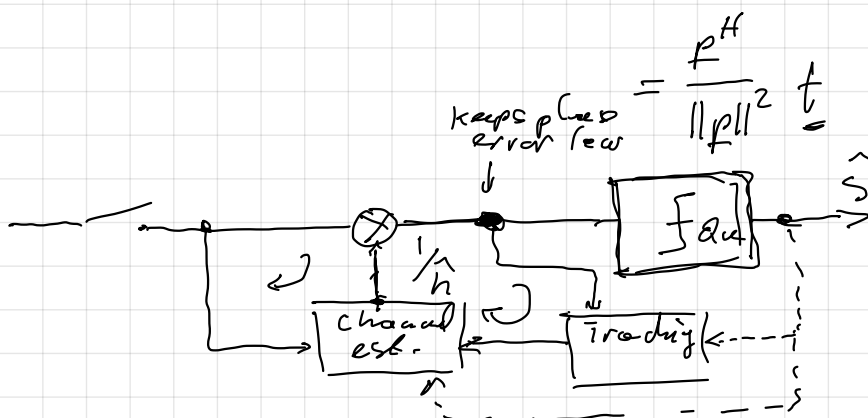
$$\hat{s} = \text{Qu}\left(\frac{1}{\underline{h}} \underline{y}\right) = \text{Qu}\left(\frac{\underline{h}^*}{\|\underline{h}\|^2} \underline{y}\right) \quad \frac{1}{\underline{h}} = \frac{\underline{h}^*}{\|\underline{h}\|^2}$$

Estimate the channel:



p : pilot symbols $\rightarrow$ training samples t

$$\hat{h} = \arg \min_h \|t - hp\|^2 \rightarrow \hat{h} = \arg \min_h \|\underline{t} - p\underline{h}\|^2$$



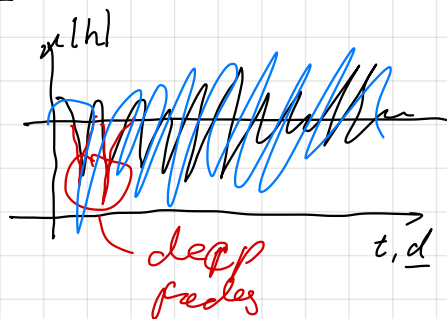
Diversity Receiver

Basic Idea: Avoid deep fades

→ try to get multiple channels

$Pr(h_1 < \gamma) \rightarrow \text{deep fade}$

$Pr(h_2 < \gamma) \rightarrow \text{deep fade}$ } h_1, h_2 independent



$$Pr(|h_1| < \gamma \text{ and } |h_2| < \gamma) = \underbrace{Pr(|h_1| < \gamma)}_{< 1} \underbrace{Pr(|h_2| < \gamma)}_{< 1}$$

Multiple channel realizations reduces the risk of a deep fade everywhere

⇒ Diversity

Forms of diversity:

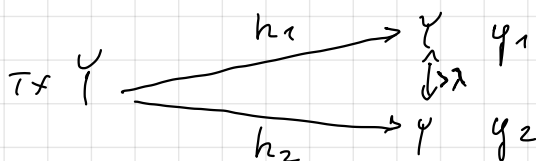
• Time diversity

• Frequency diversity

• Space → multi-antenna receiver



Example: Space (Rx-Diversity)



h_1, h_2 : independent

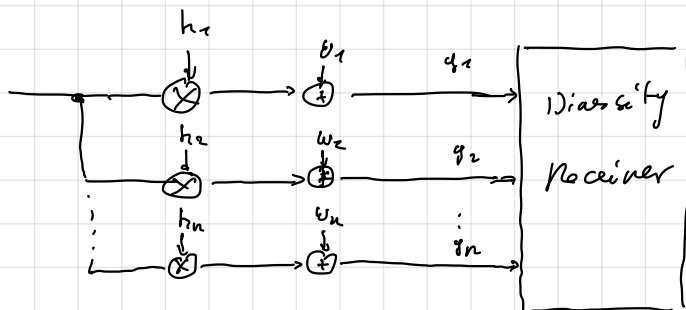
$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_1 \\ h_2 \end{bmatrix} x + \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$$

In vector notation $y = \underline{h}x + \underline{w}$

How to combine y_1 and y_2 ?

→ Selection diversity

↳ leaves the worse branch
↳ not very optimal



ML Receiver

$y_e = h_e s + w_e \rightarrow$ collect in vector notation

L : Branches $\underline{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_L \end{bmatrix}$ $\underline{h} = \begin{bmatrix} h_1 \\ \vdots \\ h_L \end{bmatrix}$ $\underline{w} = \begin{bmatrix} w_1 \\ \vdots \\ w_L \end{bmatrix} \in \mathcal{CW}(0, \underline{I}, \sigma_n^2)$
i.i.d. compl. Gaussian

ML criterion

$$\hat{s} = \arg \min_s \exp \left(- \underbrace{\|y - \underline{h}s\|^2}_{\|y - \underline{h}s\|^2} \right)$$

$$\mathcal{L}(s) = -\|y\|^2 + 2 \operatorname{Re} \{ s^* \underline{h}^H y \} - \|\underline{h}s\|^2$$

• Remove $-\|y\|^2$

$$\rightarrow \arg \min \underbrace{\|\underline{h}\|^2 \|s\|^2 - 2 \operatorname{Re} \{ s^* \underline{h}^H y \}}_{\rightarrow \text{max}}$$

max if $\underline{s}^* = \underline{y}^H \underline{h} \alpha$

find α

$$\arg \min \underbrace{\|\underline{h}\|^2 \|\underline{h}\|^2 \|y\|^2 \alpha^2 - 2 \|y\|^2 \|\underline{h}\|^2 \cdot \alpha}$$

$$\frac{d}{d\alpha} \rightarrow 0 \Rightarrow \underline{\underline{\alpha = \frac{1}{\|\underline{h}\|^2}}}$$

$$\hat{s} = \frac{\underline{h}^H}{\|\underline{h}\|^2} y \rightarrow \text{Matched filter}$$

BER:

$$\tilde{s} = \frac{h^H}{\|h\|^2} y = \frac{h^H}{\|h\|^2} (hs + w)$$

$$y = \frac{h}{r} s + w$$

$$E_h \left\{ Q \left(\sqrt{\|h\|^2 \text{SNR}} \right) \right\}$$

Note $\|h\|^2 = \sum_{e=1}^L |h_e|^2$

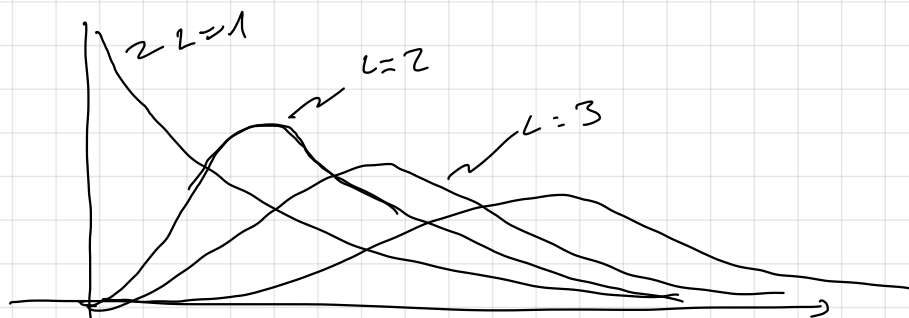


Sum of Squared Gaussians

Distribution of $\|h\|^2$: Chi-Squared χ_{2L}

with $2L$ degrees of freedom

$$f(\|h\|^2) = \frac{1}{(L-1)!} e^{-\|h\|^2} (\|h\|^2)^{L-1}$$

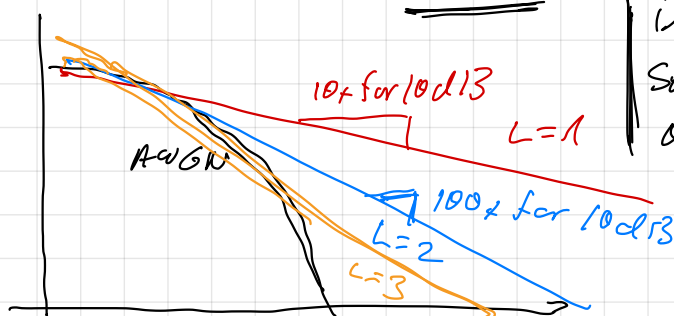


Outage probability:

$$\Pr(\|h\|^2 \text{SNR} < \gamma) = \int_0^{\gamma/\text{SNR}} f(\|h\|^2) d\|h\|^2$$

$$\approx \frac{1}{L!} \cdot \frac{1}{\text{SNR}^L}$$

Outage $\approx$ BER decays with $\left(\frac{1}{\text{SNR}} \right)^L$



Diversity Order: Steepness of BER/Outage for $\text{SNR} \rightarrow \infty$